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MAT251_TW | Chapter 7: Sequences and Series

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Question **1**

Complete

Not graded

Which of the following is the sum of the series $\sum_{n=0}^{\infty}\left(\frac{1}{\sqrt{2}}\right)^n$?

Select one:

- a. $1-\sqrt{2}$
- b. $-\sqrt{2}$
- c. $\frac{\sqrt{2}}{1-\sqrt{2}}$
- d. $2+\sqrt{2}$

$$\begin{aligned}\sum_{n=0}^{\infty}\left(\frac{1}{\sqrt{2}}\right)^n &= \frac{1}{1-\frac{1}{\sqrt{2}}}\\ &= \frac{1}{\frac{\sqrt{2}-1}{\sqrt{2}}}\\ &= \frac{\sqrt{2}}{\sqrt{2}-1}\\ &= \frac{\sqrt{2}}{\sqrt{2}-1}\cdot\frac{\sqrt{2}+1}{\sqrt{2}+1}\\ &= \frac{\sqrt{2}\left(\sqrt{2}+1\right)}{2-1}\\ &= 2+\sqrt{2}\end{aligned}$$

The correct answer is: $2+\sqrt{2}$

Question **2**

Complete

Not graded

Which of the following is the sum of the series $\sum_{n=0}^{\infty}\left(\frac{5}{2^n}+\frac{1}{3^n}\right)$?

Select one:

- a. 10
- b. 3/2
- c. 13/2
- d. 23/2

$$\sum_{n=0}^{\infty}\left(\frac{5}{2^n}+\frac{1}{3^n}\right)=\sum_{n=0}^{\infty}\left(\frac{5}{2^n}\right)+\sum_{n=0}^{\infty}\left(\frac{1}{3^n}\right)$$

$$=\sum_{n=0}^{\infty}5\left(\frac{1}{2}\right)^n+\sum_{n=0}^{\infty}\left(\frac{1}{3}\right)^n$$

$$=\frac{5}{1-\frac{1}{2}}+\frac{1}{1-\frac{1}{3}}$$

$$=\frac{5}{\frac{1}{2}}+\frac{1}{\frac{2}{3}}$$

$$=10+\frac{3}{2}$$

$$=\frac{20}{2}+\frac{3}{2}$$

$$=\frac{23}{2}$$

The correct answer is: 23/2

Question **3**

Complete

Not graded

The infinite series $\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \ldots$ can be written in which of the following forms?

Select one:

- a. $\sum_{k=1}^{\infty} \frac{1}{k}$
- b. $\sum_{k=1}^{\infty} \frac{1}{k+3}$
- c. $\sum_{k=1}^{3} \frac{1}{k}$
- d. $\sum_{k=0}^{\infty} \frac{1}{k+3}$

The numerator is always 1, and the denominators start at 3 and increase by 1.

$$\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\cdots=\sum_{n=3}^{\infty}\frac{1}{n}$$

$$=\sum_{k=0}^{\infty}\frac{1}{k+3}$$

Notice the use of the substitution $n = k + 3$.

The correct answer is: $\sum_{k=0}^{\infty} \frac{1}{k+3}$

Question **4**

Complete

Not graded

Which of the following is the sum of

$2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \ldots + \frac{2}{3^n} + \ldots .?$

Select one:

- a. ∞
- b. 3
- c. 1/3
- d. 6

$$\sum_{n=0}^{\infty} 2\left(\frac{1}{3}\right)^n = \frac{2}{1-(1/3)}$$
$$= \frac{2}{2/3}$$
$$= (2)\left(\frac{3}{2}\right)$$
$$= 3$$

The correct answer is: 3

Question **5**

Complete

Not graded

Use this notation for the questions in this exercise.

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \ldots \Rightarrow$$

$$S_m = a_1 + a_2 + a_3 + \ldots + a_m + a_{m+1}$$

Which of the following is the sum of the telescoping series $\sum_{k=1}^{\infty} ln\left(1 + \frac{1}{k}\right)$?

Select one:

- a. 2
- b. ln(2)
- c. 0
- d. It does not exist (the series diverges).

$$\sum_{k=1}^{\infty} \ln\left(1 + \frac{1}{k}\right) = \lim_{m \to \infty} \left[\ln(m+2)\right]$$

$$= +\infty$$

Therefore, the series diverges or it does not have a limit.

The correct answer is: It does not exist (the series diverges).

Question **6**

Complete

Not graded

Use this notation for the questions in this exercise.

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \ldots \Rightarrow$$

$$S_m = a_1 + a_2 + a_3 + \ldots + a_m + a_{m+1}$$

Find constants A and B such that

$$\frac{k-1}{2^{k+1}} = \frac{Ak}{2^K} - \frac{B(k+1)}{2^{k+1}}$$

Select one:

- a. $A = 1, B = 1$
- b. $A = 0, B = 1$
- c. $A = 1, B = -1$
- d. $A = 0, B = -1$

$$\frac{k-1}{2^{k+1}} = \frac{Ak}{2^k} - \frac{B(k+1)}{2^{k+1}}$$

$$= \frac{2Ak - B(k+1)}{2^{k+1}}$$

$$= \frac{(2A - B)k - B}{2^{k+1}} \Leftrightarrow -B = -1$$

and

$$2A - B = 1 \Leftrightarrow A = 1, B = 1$$

The correct answer is: $A = 1, B = 1$

Question **7**

Complete

Not graded

Use a geometric series to determine which of the following fractions is equal to $0.121\overline{212}$.

Select one:

- a. $\frac{3}{25}$
- b. $\frac{4}{33}$
- c. $\frac{121}{1000}$
- d. $\frac{2}{15}$
- e. None of the above

Factor out the repeating 12 to express the decimal in terms of a geometric series.

$0.1212\overline{12}$

$= 12\left(0.010\overline{101}\right)$

$= 12\left(\frac{1}{100}\right) + 12\left(\frac{1}{100}\right)^2 + 12\left(\frac{1}{100}\right)^3 + \cdots$

$= \sum_{n=1}^{\infty} 12\left(\frac{1}{100}\right)^n$

$= \sum_{n=0}^{\infty} \frac{12}{100}\left(\frac{1}{100}\right)^n$

$= \sum_{n=0}^{\infty} \frac{3}{25}\left(\frac{1}{100}\right)^n$

This is a geometric series of the form $\sum_{n=0}^{\infty} ar^n$, where $a = \frac{3}{25}$ and $r = \frac{1}{100}$.

Since $|r| < 1$, the identity for geometric series, $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$ may be applied:

$$0.1212\overline{12} = \sum_{n=0}^{\infty} \frac{3}{25}\left(\frac{1}{100}\right)^n$$

$$= \frac{3/25}{1 - 1/100}$$

$$= \frac{3}{25} \cdot \frac{100}{99} = \frac{4}{33}$$

The correct answer is: $\frac{4}{33}$

Question **8**

Complete

Not graded

Evaluate $\sum_{n=0}^{\infty}(-3)^{1-n}$.

Select one:

- a. −4
- b. −9/4
- c. −1/12
- d. −9/2

$$\sum_{n=0}^{\infty} -3\left(\frac{-1}{3}\right)^n = \lim_{n\to\infty} S_n$$
$$= \frac{-3}{1-\left(\frac{-1}{3}\right)}$$
$$= \frac{-3}{4/3}$$
$$= -\frac{9}{4}$$

The correct answer is: −9/4

Question **9**

Complete

Not graded

Use this notation for the questions in this exercise.

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \ldots \Rightarrow$$

$$S_m = a_1 + a_2 + a_3 + \ldots + a_m + a_{m+1}$$

Evaluate $\frac{4}{1\times 2} + \frac{4}{2\times 3} + \frac{4}{3\times 4} + \ldots$

Select one:

- a. 4
- b. 2
- c. It does not converge.
- d. 0

$$\frac{4}{1\times 2} + \frac{4}{1\times 2} + \frac{4}{1\times 2} + \cdots = \sum_{k=1}^{\infty} \frac{4}{k(k+1)}$$

$$= \lim_{m\to\infty} 4\left(1 - \frac{1}{m+2}\right)$$

$$= 4(1-0)$$

$$= 4$$

The correct answer is: 4

Question **10**

Complete

Not graded

Which of the following is the sum of the series $\sum_{n=0}^{\infty}\left(\frac{2^n}{5^n}\right)$?

Select one:

- a. 5/3
- b. 3/5
- c. 5/2
- d. 10

$$\sum_{n=0}^{\infty}\left(\frac{2}{5}\right)^n = \frac{1}{1-\frac{2}{5}}$$
$$= \frac{1}{\frac{3}{5}}$$
$$= 1 \cdot \frac{5}{3}$$
$$= \frac{5}{3}$$

The correct answer is: 5/3

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